

Git Reset Command

Introduction

The `git reset` command is used to move the current branch (HEAD) to a previous commit. It can also be used to unstage files or remove changes depending on the option used.

Unlike `git revert`, `git reset` can rewrite commit history, so it should be used carefully.

 **Watch the Video:**

https://youtu.be/qL57wSk9UVw?si=__eIH1lrNeil5FJ

What You Will Learn

- What Git Reset is and how it works
- Understanding `--soft`, `--mixed`, and `--hard` reset options
- Difference between Git Reset and Git Revert
- Practical use of `HEAD~1` to undo commits
- Real-world scenarios of using reset effectively

Git Reset Command Notes

Introduction

The `git reset` command is used to move the current branch (HEAD) to a previous commit. It helps in undoing commits, unstaging files, or removing changes depending on the option used.

It is a powerful command because it can **modify commit history**, so it should be used carefully.

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Types of Git Reset

1. Soft Reset

```
git reset --soft <commit>
```

- Moves HEAD to a previous commit
- Keeps changes staged (in index)
- Does not remove working directory changes

Use case: Undo commit but keep changes ready for recommit.

2. Mixed Reset (Default)

```
git reset --mixed <commit>
```

or

```
git reset <commit>
```

- Moves HEAD to previous commit
- Unstages files
- Keeps changes in working directory

Use case: Undo staging but keep code changes.

3. Hard Reset

```
git reset --hard <commit>
```

- Moves HEAD to previous commit
- Removes staged changes
- Deletes all working directory changes

Use case: Completely discard changes

Warning: Data is permanently lost if not committed.

Common Example

```
git reset HEAD~1
```

- Moves branch back by 1 commit
- Default is mixed reset

Git Reset vs Git Revert

Git Reset	Git Revert
Deletes/rewrites history	Keeps history
Local use recommended	Safe for shared repos
Can remove commits	Creates new commit

Key Points

- HEAD represents the current position in Git history
- `reset` moves HEAD to a different commit
- Use `soft` , `mixed` , or `hard` depending on need
- Avoid using reset on shared/public branches
- Prefer `revert` for already pushed commits